TCS TECHNOLOGY DAY 2026 · BANKING: AUTOMATED LOAN ELIGIBILITY CHATBOT

LoanMitra

Model evaluation report: accuracy, latency and robustness

Measured 2026-10-28 · code version e65daf1 · all numbers produced by scripts in the repository

100%
Verdict accuracy, rules
36/36 conversations

97.2%
Verdict accuracy, NVIDIA AI
35/36 conversations

91.7%
Verdict accuracy, OpenRouter AI
11/12 (subset)

99.6%
AI extraction accuracy
282/283 fields, first message

100%
Replies under 15 s
worst configuration

SUMMARY

What we measured and what it shows

LoanMitra is a single-agent chatbot that decides loan eligibility for personal, home, car and education loans. Its design splits the work: an **AI model understands the applicant and explains the result**, while a **deterministic rules engine makes every decision**. This report measures each part separately and the whole conversation end to end, for every AI model we use and for the no-AI fallback.

- **Brief target: 80%+ accuracy, replies under 15 s.** Rules-based agent: 36/36 verdicts correct, 69/69 replies under 15 s.
- **NVIDIA Nemotron 3 Ultra (the AI model measured live):** 35/36 verdicts correct (97.2%); 282/283 facts extracted correctly from the applicant's first message (99.6%); 32/38 model calls succeeded; replies that used the model took 3.5 s median, 13.0 s p95. The 6 failed calls (server errors and timeouts) were answered by the fallback, so no conversation broke.
- **Nemotron 3.5 Super via OpenRouter (free tier):** 11/12 verdicts correct on a 12-case subset; median reply 1.9 s.
- **Rules engine:** 17/17 boundary cases correct; 0 violations of 22 consistency rules across 20,000 random profiles; 0.01 ms per decision.
- **Robustness and privacy:** 81 real-world extraction messages, 359 edge-value inputs, 17 PII-masking cases and 14 HTTP security checks, all passing.
- **Why accuracy holds when a model is weak:** the verdict always comes from the rules engine, and the applicant confirms the extracted details before any verdict. A model error shows up as an extra question, not a wrong decision.

1 · METHOD

Test data, configurations and metrics

Test data

- **36 labelled conversations** (data/test_cases.json): 29 English, 3 Hindi, 4 Hinglish. Expected verdicts: 10 eligible, 13 conditional, 10 not eligible, 3 incomplete (the applicant doesn't know a required fact).
- **17 rule-boundary cases**: applicants placed exactly on cut-offs (credit score, FOIR, loan-to-value, age at maturity).
- **20,000 random applicant profiles** (fixed seed) for consistency checks, and **81 realistic messages** written to break the extractor (“5 lacs”, “12 LPA”, “70 hazaar”, father's income vs own income).
- Every label was set by hand before running the models. All data is synthetic.

Configurations evaluated (one backend at a time, no fallback chain)

Configuration	Model	Cases	Why
Rule-based agent (no AI)	Rule-based agent (no AI)	36 (all)	Baseline; also the live fallback
NVIDIA Nemotron 3 Ultra (550B)	nvidia/nemotron-3-ultra-550b-a55b	36 (all)	Main live AI model (free API credits)
Rule-based agent (no AI)	Rule-based agent (no AI)	12 (stratified subset)	Baseline; also the live fallback
Nemotron 3.5 Super (OpenRouter free)	nvidia/nemotron-3.5-super-vl:free	12 (stratified subset)	Default free model in the app

The 12-case subset contains every Hindi and Hinglish case plus English cases covering each verdict and loan type. It keeps the OpenRouter run inside the free tier's daily request limit; the rules agent is reported on the same subset for a like-for-like comparison.

Procedure

- The simulated applicant sends the labelled message. When the bot asks for missing facts, the applicant answers from the label (up to 8 turns); when the bot shows the details for confirmation, the applicant replies “Yes, confirm”.
- Each reply runs under the production settings: a hard 13-second turn budget. If the model cannot finish, the engine's own explanation is sent (“fast path”); if the model errors or times out, the rules agent answers (“fallback”).
- Every raw model call is timed and recorded with its outcome (success, HTTP error, timeout).

Metrics

Metric	Definition
Verdict accuracy	Final verdict (ELIGIBLE / CONDITIONAL / NOT_ELIGIBLE / INCOMPLETE) equals the label
First-message extraction	Share of labelled fields correctly captured from the applicant's first message alone
Final field accuracy	Share of labelled fields correct in the profile the verdict was based on
Latency p50 / p95 / max	Wall-clock time per reply, measured at the server
Under 15 s	Share of replies faster than the brief's 15-second target
Reply source	Model (AI wrote the reply) · fast path (engine text, AI ran out of time) · fallback (AI failed) · engine (no AI needed, e.g. confirmation)
Model call success	Raw API calls that returned a usable answer
Language match	Every reply in the applicant's script (Devanagari for Hindi, Latin otherwise)
Precision / recall / F1	Per verdict class, from the confusion matrix

2 · RESULTS

Model comparison

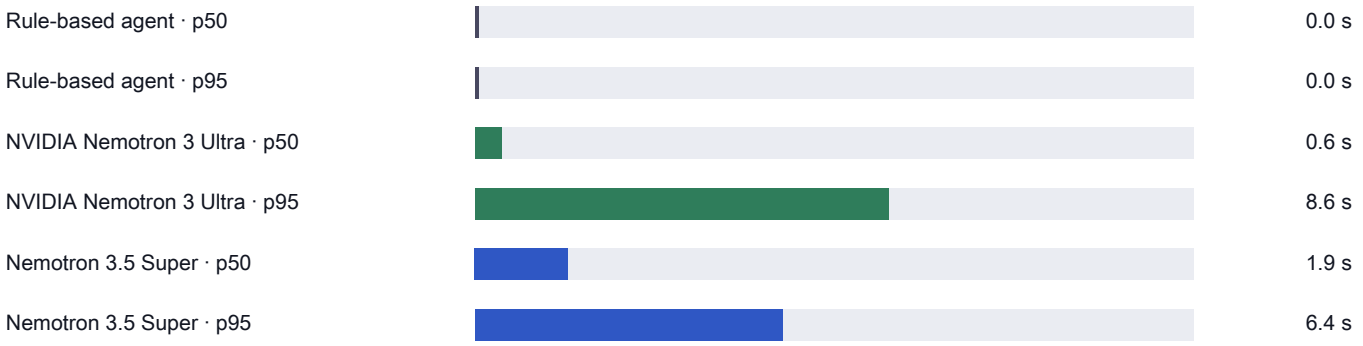
Metric	Rule-based agent all 36	NVIDIA Nemotron 3 Ultra all 36	Rule-based agent subset 12	Nemotron 3.5 Super subset 12
Verdict accuracy	36/36 (100%)	35/36 (97.2%)	12/12 (100%)	11/12 (91.7%)
First-message extraction	283/283 (100%)	282/283 (99.6%)	100/100 (100%)	99/100 (99.0%)
Final field accuracy	283/283 (100%)	283/283 (100%)	100/100 (100%)	100/100 (100%)
Avg turns to verdict	1.9	2.0	1.9	2.1
Latency p50	0.0 s	0.6 s	0.0 s	1.9 s
Latency p95	0.0 s	8.6 s	0.0 s	6.4 s
Latency max	0.0 s	13.0 s	0.0 s	9.6 s
Replies under 15 s	69/69 (100%)	71/71 (100%)	23/23 (100%)	25/25 (100%)
AI-turn latency p50 / p95	—	3.5 / 13.0 s	—	3.3 / 6.6 s
AI-turn latency max	—	13.0 s	—	9.6 s
Replies written by the model	—	3	—	1
Fast path (engine text)	—	28	—	12
Fallback to rules agent	—	6	—	0
Model call success	—	32/38 (84.2%)	—	14/14 (100%)
Model call latency p50	—	3.5 s	—	3.1 s
Language match (all cases)	36/36	35/36	12/12	12/12
Language match (Hindi + Hinglish)	7/7	6/7	7/7	7/7

Latency p50 / p95 covers every reply, including instant ones such as the confirmation step. AI-turn latency covers only replies where the model was called. “Fast path” means the model extracted the facts but the engine wrote the explanation because a second model call would not have fitted inside the 13-second budget.

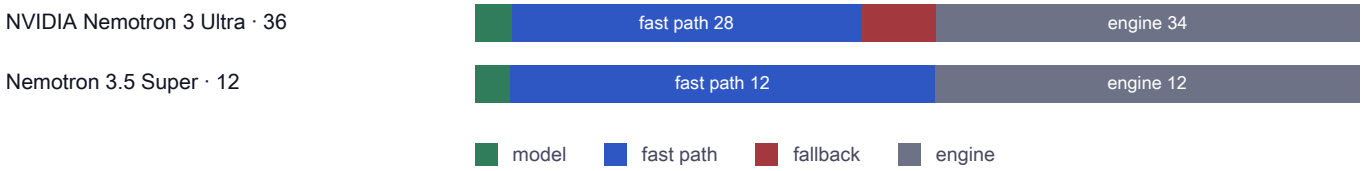
Verdict accuracy (target 80%)



Median and 95th-percentile reply time (bar scale 0–15 s)



How each reply was produced (AI configurations)



3 · PER-CLASS METRICS

Confusion matrices, precision, recall and F1

Rule-based agent (no AI) · 36 conversations

Expected \ Got	ELIGIBLE	CONDITIONAL	NOT_ELIGIBLE	INCOMPLETE	Precision	Recall	F1
ELIGIBLE	10	0	0	0	1.00	1.00	1.00
CONDITIONAL	0	13	0	0	1.00	1.00	1.00
NOT_ELIGIBLE	0	0	10	0	1.00	1.00	1.00
INCOMPLETE	0	0	0	3	1.00	1.00	1.00

NVIDIA Nemotron 3 Ultra (550B) · 36 conversations

Expected \ Got	ELIGIBLE	CONDITIONAL	NOT_ELIGIBLE	INCOMPLETE	Precision	Recall	F1
ELIGIBLE	10	0	0	0	1.00	1.00	1.00
CONDITIONAL	0	13	0	0	0.93	1.00	0.96
NOT_ELIGIBLE	0	0	10	0	1.00	1.00	1.00
INCOMPLETE	0	1	0	2	1.00	0.67	0.80

Nemotron 3.5 Super (OpenRouter free) · 12 conversations

Expected \ Got	ELIGIBLE	CONDITIONAL	NOT_ELIGIBLE	INCOMPLETE	Precision	Recall	F1
ELIGIBLE	6	0	0	0	1.00	1.00	1.00
CONDITIONAL	0	4	0	0	0.80	1.00	0.89
NOT_ELIGIBLE	0	0	1	0	1.00	1.00	1.00
INCOMPLETE	0	1	0	0	0.00	0.00	0.00

Accuracy by language and loan type

Group	Rule-based agent · 36	NVIDIA Nemotron 3 Ultra · 36	Rule-based agent · 12	Nemotron 3.5 Super · 12
English	29/29	29/29	5/5	5/5
Hindi	3/3	3/3	3/3	3/3
Hinglish	4/4	3/4	4/4	3/4
Personal loan	17/17	16/17	6/6	5/6
Home loan	6/6	6/6	3/3	3/3
Car loan	6/6	6/6	2/2	2/2
Education loan	5/5	5/5	1/1	1/1

4 · ERROR ANALYSIS

Where the models struggled

NVIDIA Nemotron 3 Ultra (550B)

Case	Lang	Expected	Got	Turns	Fields at verdict	Reply sources
nl34	hinglish	INCOMPLETE	CONDITIONAL	2	7/7	engine, fast path

Model call failures: HTTP 500 ×3; ReadTimeout ×3

Nemotron 3.5 Super (OpenRouter free) · subset

Case	Lang	Expected	Got	Turns	Fields at verdict	Reply sources
nl34	hinglish	INCOMPLETE	CONDITIONAL	2	7/7	engine, fast path

Case nl34 (missed by NVIDIA Nemotron 3 Ultra and Nemotron 3.5 Super) is not a hallucination. The applicant wrote “CIBIL pata nahi” (“I don't know my credit score”). The model recorded the score as unknown, a value LoanMitra supports by design: the engine then gives a conditional verdict with a caveat. The label expected the bot to keep asking for the score. Both AI models made the same, consistent choice. Both behaviours are defensible; we count it as a miss.

A model failure never produces an unsafe verdict: when a call errors or exceeds the budget, the turn is answered by the rules agent and the conversation continues. The cost of a weak model is latency or an extra question, which is why verdict accuracy stays high even when model call success is below 100%.

5 · RULES ENGINE AND ROBUSTNESS

The parts that make every decision

17/17
Boundary cases
rules engine

20k
Random profiles
0 crashes

0
Invariant violations
239,601 checks

0.01 ms
Per decision
rules engine speed

Consistency rules checked on 20,000 random profiles

Invariant	What it guarantees	Checked	Violations
I1	EMI matches the standard formula	20,000	0
I2	FOIR = (existing EMIs + new EMI) ÷ income	20,000	0
I2b	Displayed FOIR equals the checked FOIR	20,000	0
I2c	Displayed LTV equals the checked LTV	5,109	0
I3	Verdict follows from the individual rule results	20,000	0
I4	More income never worsens the verdict	20,000	0
I5	Higher credit score never worsens it	20,000	0
I6	Fewer existing EMIs never worsen it	20,000	0
I7	A smaller amount never worsens it	16,633	0
I8	A more valuable asset never worsens it	10,159	0
I9a	A longer tenure never worsens it	8,892	0
I9b	A longer tenure never lowers the maximum amount	8,892	0
I10	The quoted maximum amount is itself eligible	5,225	0
I11	Every “eligible up to X” in a summary is true	2,412	0
I12	“Extend the tenure” advice works when applied	1,188	0
I13	“Close EMIs” advice works when applied	1,237	0
I14	“Add a co-applicant” advice works when applied	350	0
I15	“You qualify for up to X” is true	6,617	0
I16	“Reduce the tenure” advice is valid	2,482	0
I17	“Improve your score” advice works	5,464	0
I18b	LTV-capped “request up to” amount is eligible	4,941	0
I19	Eligible applicants never get a maximum below their request	20,000	0

Robustness, privacy and security suites

Suite	Cases	Failing	What it covers
Real-world extraction	81	0	Indian amounts, annual vs monthly income, co-applicant vs own facts, negations
Edge values	359	0	NaN, infinity, zero, negative, text in number fields, wrong types
PII masking	17	0	PAN, Aadhaar (checksum), phone, email, card (Luhn), account numbers; no false hits on amounts
HTTP hardening	14	0	Rate limits, body limits, schema validation, consent, security headers, session IDs

Bank-statement analysis against ground truth

Statement	Transactions (true / read)	Monthly income (true / found)	Monthly EMIs (true / found)
Demo statement (PDF)	102 / 102	Rs 82,000 / Rs 82,000	Rs 9,300 / Rs 9,300
Sample statement (PDF)	67 / 67	Rs 65,000 / Rs 65,000	Rs 8,500 / Rs 8,500
Sample statement (CSV)	67 / 67	Rs 65,000 / Rs 65,000	Rs 8,500 / Rs 8,500

6 · ISSUES FOUND AND FIXED

What testing changed in the product

Issue	Found by	Fix	Status
Applicant answered 7 questions with 7 bare values, one per line; the model was slow, the fallback could not place bare numbers, so 6 values were lost and asked again	Live demo test	List answers are now mapped in order to the details still missing, deterministically and instantly; the confirmation card shows every value with its label	Fixed, regression test added
Model asked for 7 details in one message	Live demo test	Prompt now requires the tool's order and at most 2 details per message; list mapping covers models that ignore it	Fixed
NVIDIA API: 3 server errors (HTTP 500) and 3 timeouts in 38 calls	This evaluation	Already handled: the rules agent answers that turn and the conversation continues	Working as designed
OpenRouter free tier rejected every call (daily limit used up)	This evaluation	Provider chain falls through to NVIDIA, then to the rules agent; for a production deployment use a paid key	Mitigated
Typed Marathi detected as Hindi (same script)	Live demo test	Marathi-only words now distinguish the two languages	Fixed

7 · LIMITATIONS

How to read these numbers

- **Same-team test data.** The labelled conversations were written by the team that built the system. They show the system is correct and consistent on known phrasing; they are not a blind test. Next step: 25 messages written independently by the five team members, scored without changes to the code.
- **Small samples.** 36 labelled conversations give wide confidence intervals; one miss moves accuracy by about 3 points.
- **Free-tier models.** Latency and error rates depend on shared free infrastructure at the time of the run, and will differ on a dedicated deployment.
- **Regional languages.** Bengali, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Punjabi and Odia go through machine translation and were not part of this labelled set.
- **Synthetic policy.** Rules and lenders are illustrative; the evaluation shows the engine applies its policy correctly, not that the policy matches any real bank.

APPENDIX

Per-case results · NVIDIA Nemotron 3 Ultra (550B)

Case	Lang	Loan	Expected	Got	OK	Turns	Max reply	Sources
nl01	en	personal	ELIGIBLE	ELIGIBLE	✓	2	0.6 s	engine, fallback
nl02	en	home	CONDITIONAL	CONDITIONAL	✓	2	13.0 s	engine, fallback
nl03	hinglish	auto	ELIGIBLE	ELIGIBLE	✓	2	6.1 s	engine, fast path
nl04	hi	education	CONDITIONAL	CONDITIONAL	✓	3	7.7 s	engine, fallback, model
nl05	en	personal	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	4.7 s	engine, fast path
nl06	en	personal	CONDITIONAL	CONDITIONAL	✓	2	8.6 s	engine, fast path
nl07	en	personal	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	6.0 s	engine, fast path
nl08	en	personal	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	0.6 s	engine, fallback
nl09	hinglish	personal	ELIGIBLE	ELIGIBLE	✓	2	2.5 s	engine, fast path
nl10	hi	personal	CONDITIONAL	CONDITIONAL	✓	2	2.1 s	engine, fast path
nl11	en	home	ELIGIBLE	ELIGIBLE	✓	2	3.0 s	engine, fast path
nl12	en	home	CONDITIONAL	CONDITIONAL	✓	2	2.7 s	engine, fast path
nl13	en	home	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	5.0 s	engine, fast path
nl14	en	auto	CONDITIONAL	CONDITIONAL	✓	2	8.2 s	engine, fast path
nl15	en	auto	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	2.1 s	engine, fast path
nl16	en	education	ELIGIBLE	ELIGIBLE	✓	2	13.0 s	engine, fallback
nl17	en	education	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	4.8 s	engine, fast path
nl18	en	education	CONDITIONAL	CONDITIONAL	✓	2	3.2 s	engine, fast path
nl19	en	—	INCOMPLETE	INCOMPLETE	✓	1	1.1 s	model
nl20	en	—	INCOMPLETE	INCOMPLETE	✓	1	3.5 s	model
nl21	en	personal	ELIGIBLE	ELIGIBLE	✓	2	13.0 s	engine, fallback
nl22	en	personal	CONDITIONAL	CONDITIONAL	✓	2	3.4 s	engine, fast path
nl23	en	personal	CONDITIONAL	CONDITIONAL	✓	2	11.1 s	engine, fast path
nl24	en	personal	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	5.3 s	engine, fast path
nl25	hinglish	home	ELIGIBLE	ELIGIBLE	✓	2	3.7 s	engine, fast path
nl26	hi	auto	ELIGIBLE	ELIGIBLE	✓	2	3.2 s	engine, fast path
nl27	en	education	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	2.9 s	engine, fast path
nl28	en	personal	ELIGIBLE	ELIGIBLE	✓	2	2.2 s	engine, fast path
nl29	en	home	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	4.3 s	engine, fast path
nl30	en	personal	CONDITIONAL	CONDITIONAL	✓	2	7.2 s	engine, fast path
nl31	en	auto	NOT_ELIGIBLE	NOT_ELIGIBLE	✓	2	3.0 s	engine, fast path
nl32	en	personal	ELIGIBLE	ELIGIBLE	✓	2	8.1 s	engine, fast path
nl33	en	personal	CONDITIONAL	CONDITIONAL	✓	2	2.4 s	engine, fast path

Case	Lang	Loan	Expected	Got	OK	Turns	Max reply	Sources
nl34	hinglish	personal	INCOMPLETE	CONDITIONAL	✗	2	8.3 s	engine, fast path
nl35	en	auto	CONDITIONAL	CONDITIONAL	✓	2	2.8 s	engine, fast path
nl36	en	personal	CONDITIONAL	CONDITIONAL	✓	2	2.6 s	engine, fast path